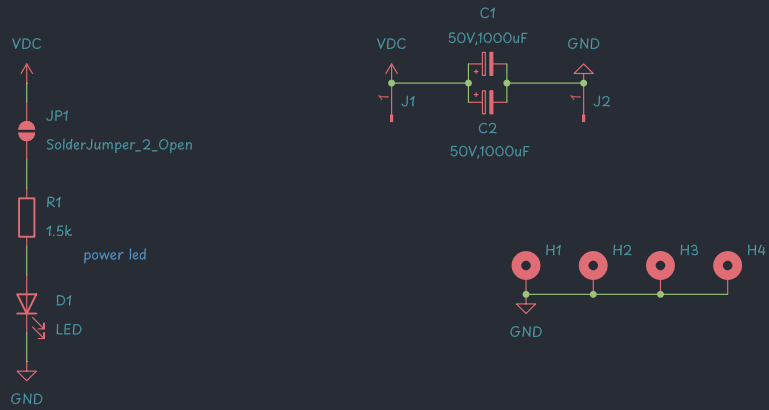
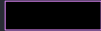


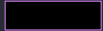


rp2040



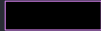
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3.3V Buck Regulator



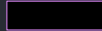
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5V Buck Regulator



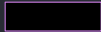
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Neopixels



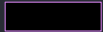
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USB 1



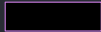
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USB 2



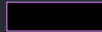
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USB 3



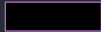
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USB 4

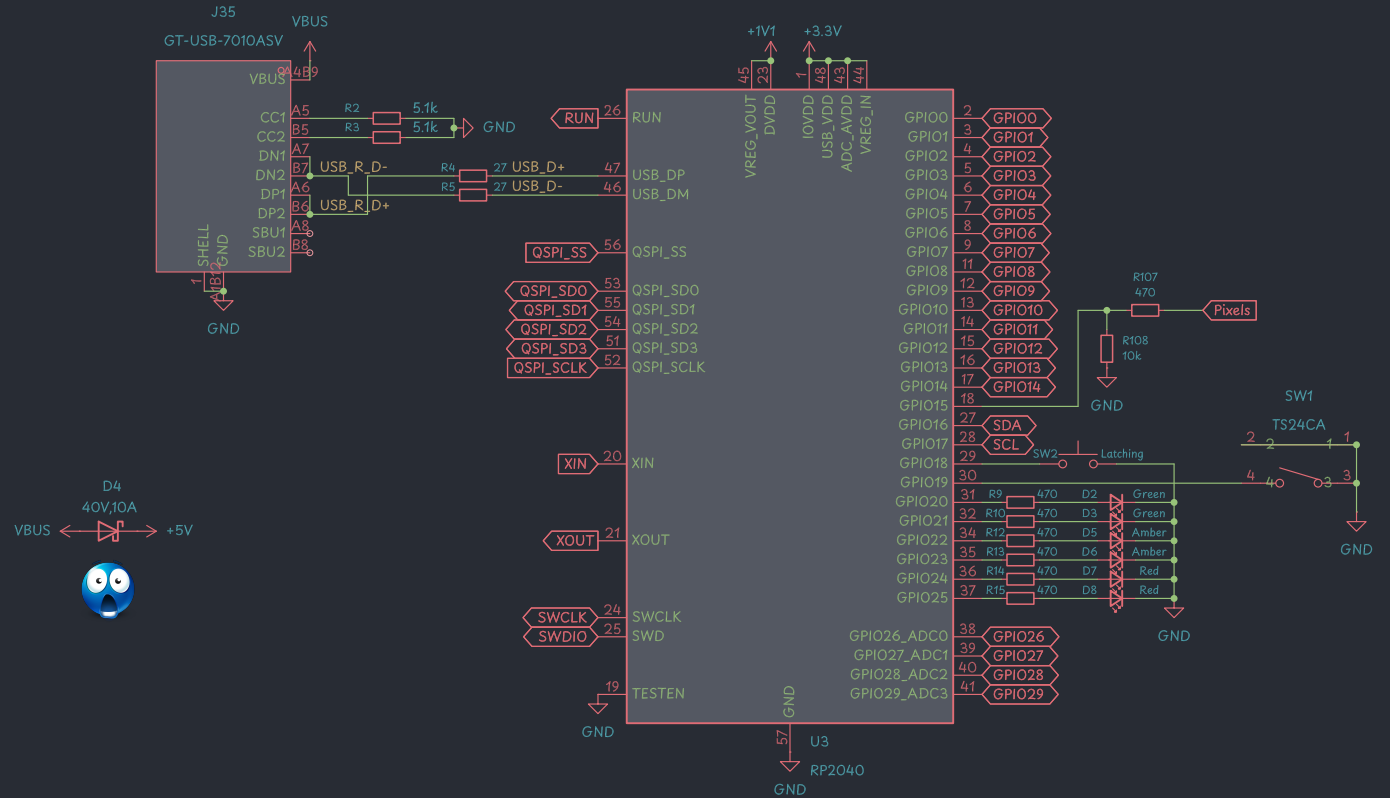
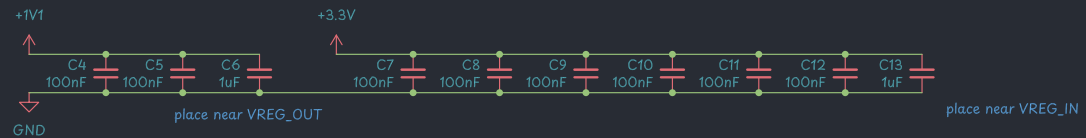
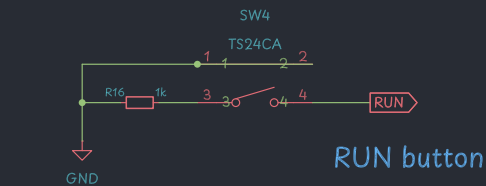
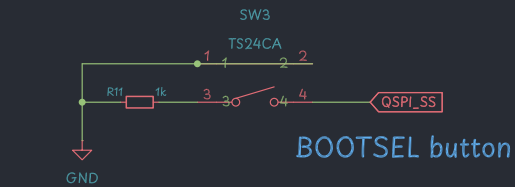
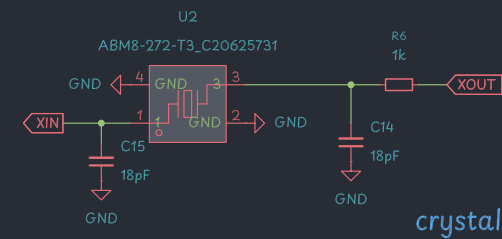
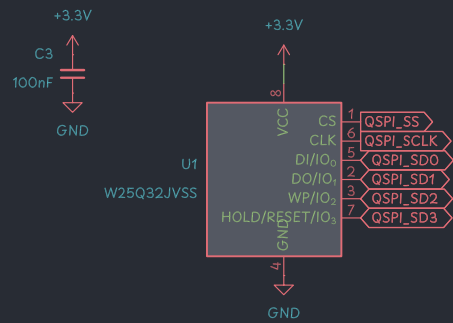


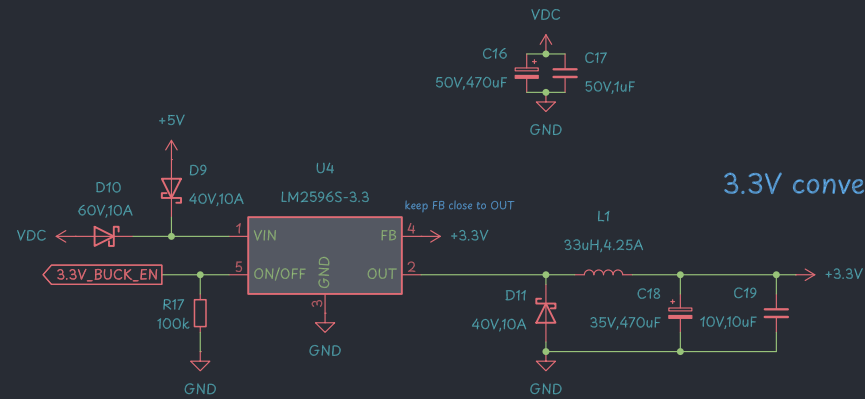
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USB 5

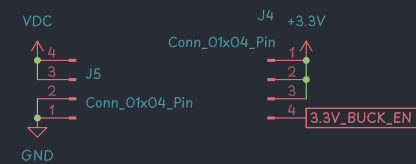


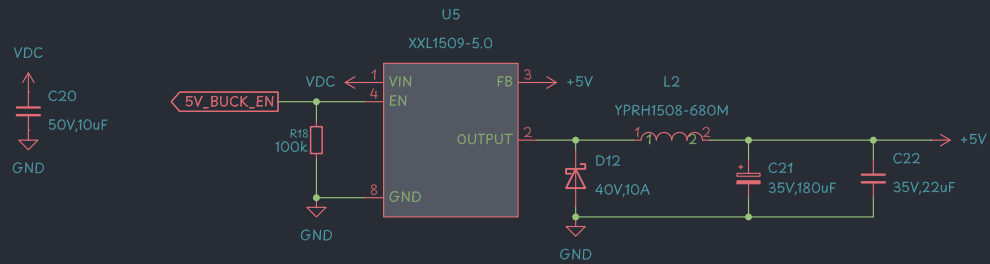
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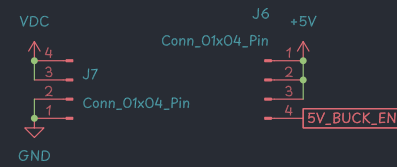


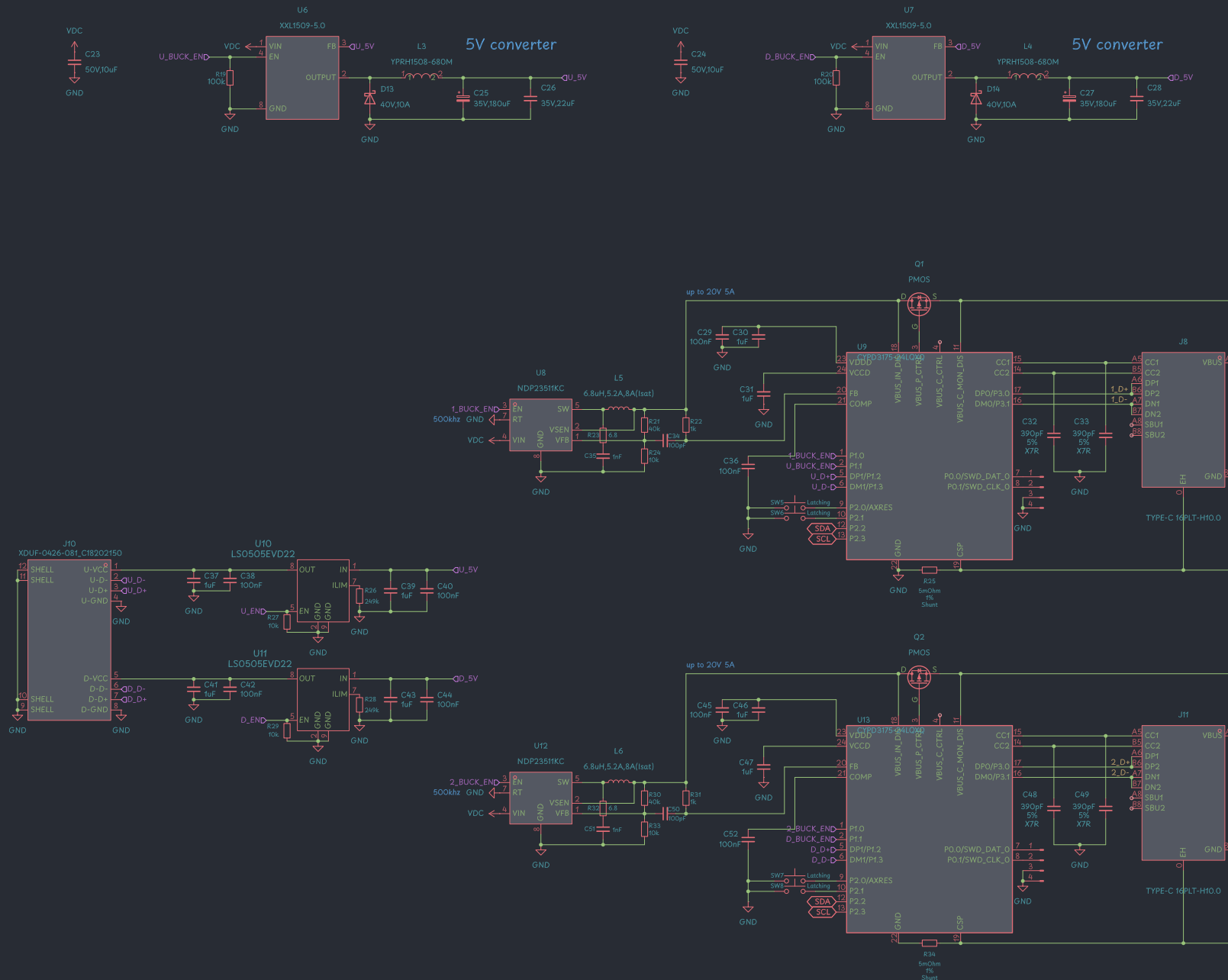
3.3V converter





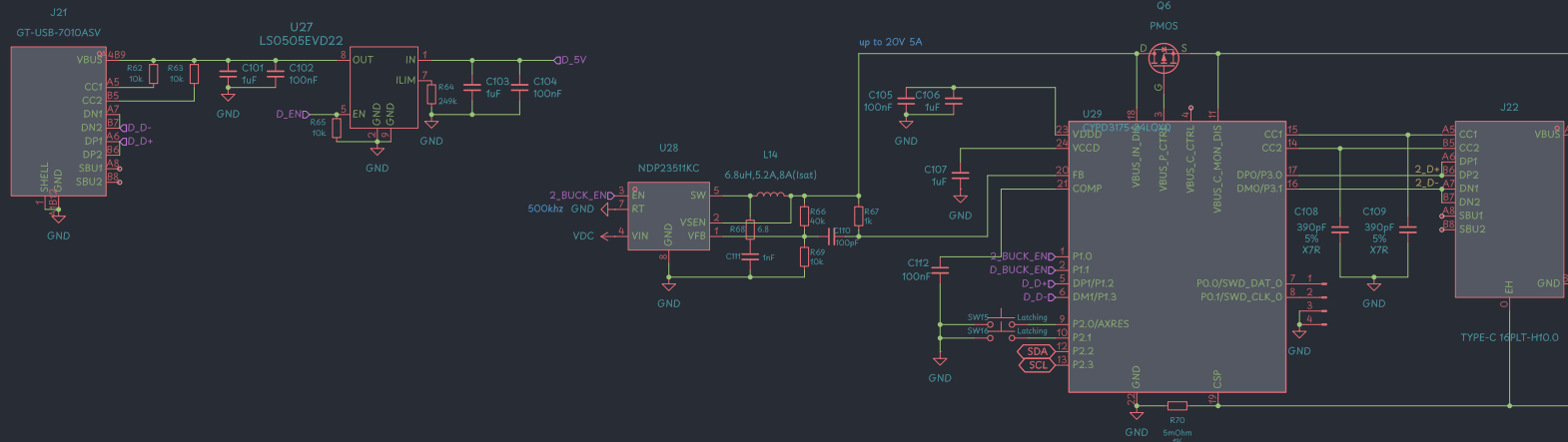
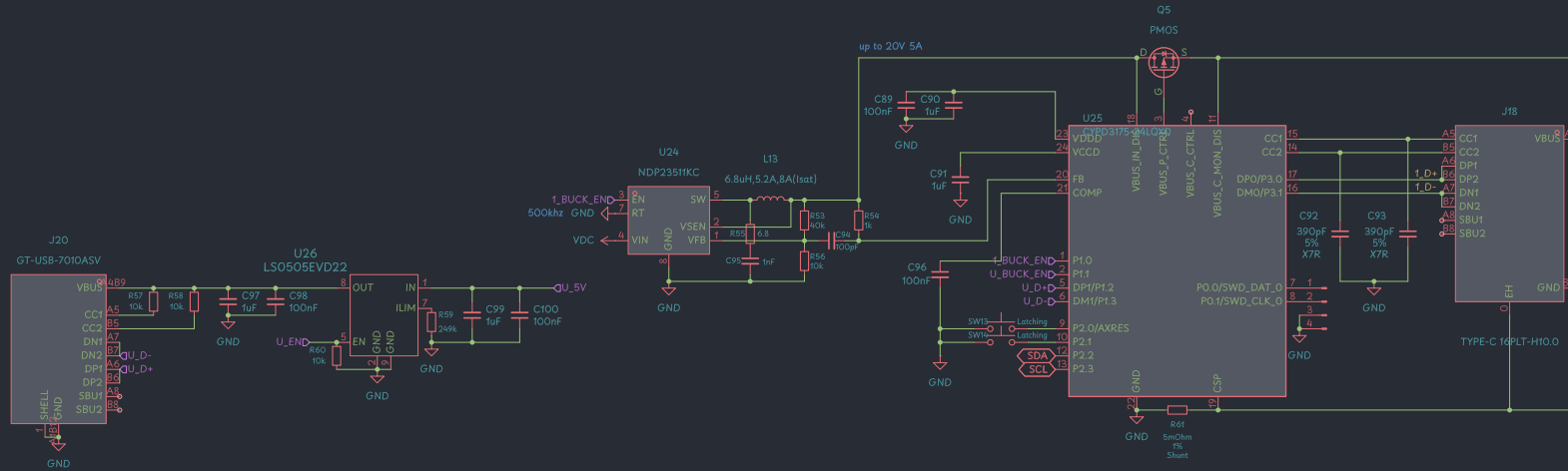
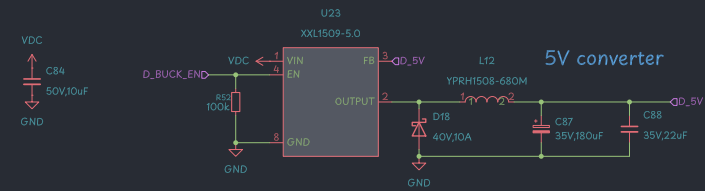
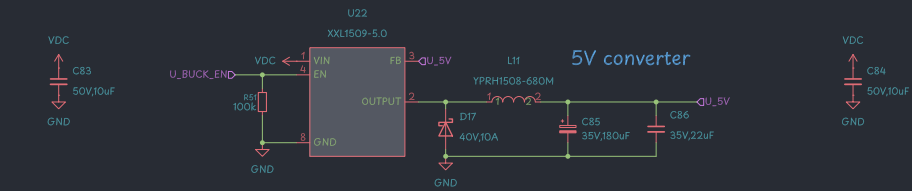
5V converter



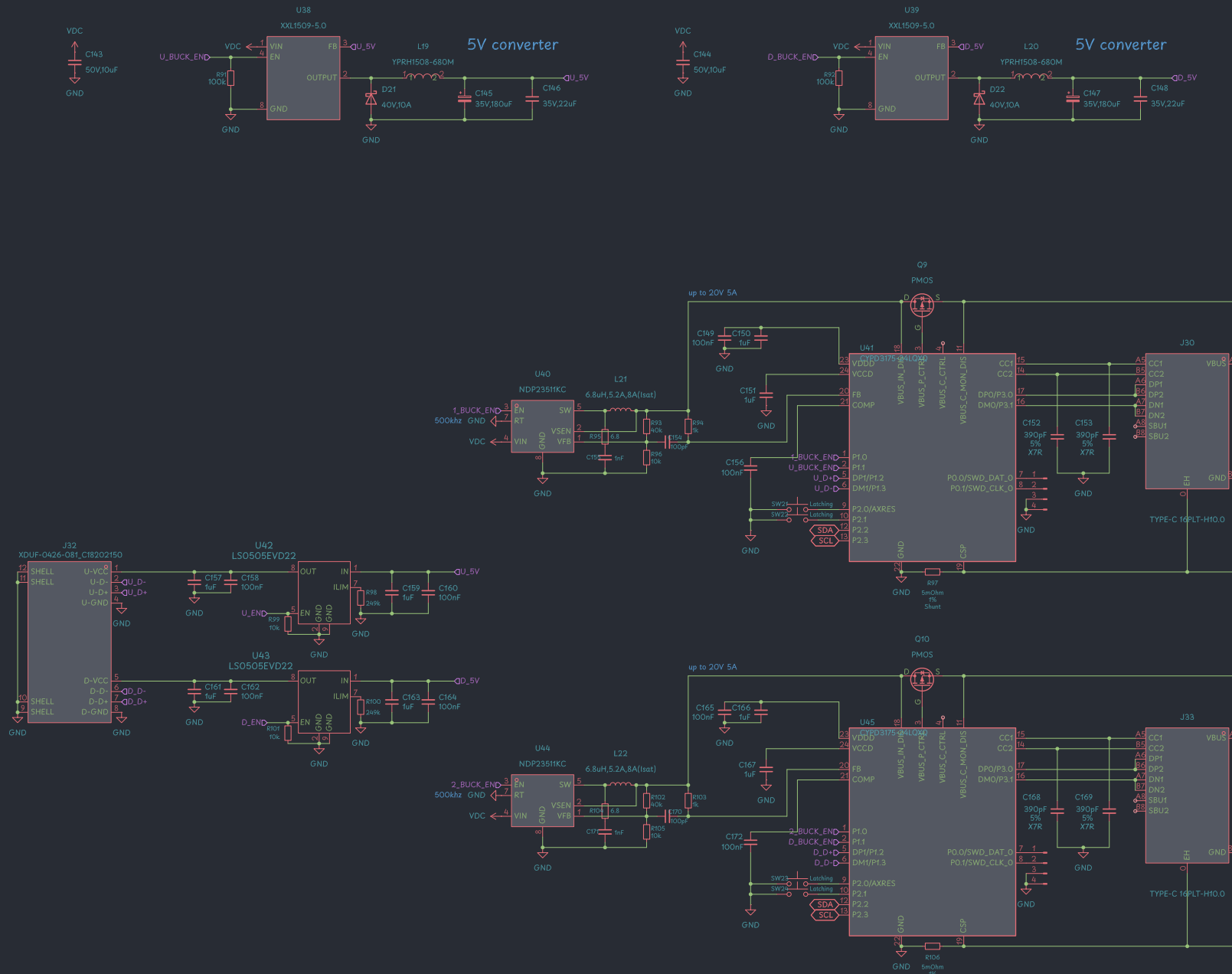


It's crucial how this is connected.
For accurate low-side sensing:

- The main VBUS supply current to the Type-C port should return to the power supply ground through this shunt resistor.
- The CSP pin senses the voltage at the "load side" of the shunt (closer to where the current enters the shunt from the Type-C VBUS ground return).
- The "system ground" or "power ground" reference for the CYPD3175's CSA would be at the other end of the shunt resistor.



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